

Profile Summary

AI & Data Science undergraduate with strong foundations in machine learning, computer vision, and web development, complemented by hands-on project experience across real-world applications. Skilled in building end-to-end solutions, including ML models, OpenCV-based systems, and cloud-integrated applications. Published researcher in Conversational AI, with exposure to NLP, transformers, and data-driven analysis. Proficient in Python, JavaScript, and tools like Power BI, with a keen interest in developing intelligent, scalable AI solutions for real-world impact.

Professional Skill

- Programming: Python, JavaScript, Web Development Fundamental (HTML, CSS)
- AI/ML & Computer Vision: Machine Learning, Artificial Intelligence, OpenCV
- Data & Analytics: Data Structures, Data Analysis, Power BI
- Web Development: HTML, CSS, DOM Manipulation
- Tools: Git, GitHub, VS Code
- Core Competencies: OOP, Problem Solving, API Integration, Debugging

2023 - Present

Education

Bachelor of Engineering - Mumbai University - Artificial Intelligence and Data Science

Projects

CS50 Python Programming | Harvard University

- Completed 8 weeks of hands-on programming assignments covering core Python concepts, including conditionals, loops, functions, file handling, and error handling
- Developed multiple mini-projects such as Tip Calculator, Math Interpreter, Fuel Gauge, Guessing Game, and Bitcoin Price Index, demonstrating problem-solving and algorithmic thinking
- Implemented real-world applications involving data validation, string manipulation, and user input handling
- Applied regular expressions, file I/O, and API integration (e.g., Bitcoin Price Index) to build practical utilities
- Practiced test-driven development and debugging techniques through structured problem sets
- Strengthened coding skills by building modular, efficient, and readable Python programs across progressive difficulty levels

Web Development & Python

- Developed 25+ projects spanning web development and Python programming, including applications, games, and utility tools
- Built full-stack web applications such as E-Commerce Website, Grocery Website, Library Management System, and Task Manager with Authentication
- Designed interactive browser-based games like Flappy Bird, Snake Game, Tic Tac Toe, and Memory Card Game using JavaScript
- Implemented utility-based applications, including Expense Tracker, Weather App, Currency & Unit Converter, and Calculator
- Applied concepts of DOM manipulation, event handling, and responsive UI design in multiple frontend projects
- Integrated core programming concepts such as logic building, data structures, and modular coding across projects
- Developed authentication-based and data-driven applications demonstrating real-world problem-solving skills
- Created multiple portfolio websites showcasing projects and improving UI/UX design skills

Data Science Projects (Power BI)

- Performed data analysis and built interactive dashboards using Power BI to extract insights from real-world datasets, focusing on pattern recognition, sentiment analysis, and predictive modeling
- Road Accident Analysis (U.S.): Analyzed accident data to identify patterns based on time of day, weather conditions, and road features; visualized accident hotspots and trends using dynamic Power BI dashboards to support data-driven decision-making
- Social Media Sentiment Analysis: Explored and visualized sentiment patterns from real Twitter data to understand public opinion towards topics/brands; created dashboards highlighting trends, polarity distribution, and user perception insights
- Bank Term Deposit Prediction: Developed a Decision Tree Classifier to predict customer subscription to term deposits; analyzed key influencing factors and visualized results to support business decision-making

Computer Vision Projects (OpenCV)

- Developed multiple real-time computer vision applications using OpenCV, focusing on gesture recognition, human-computer interaction, and visual data processing
- Sign Language & Sentiment Analysis System: Built a system to recognize hand gestures for basic sign language interpretation and integrated sentiment analysis to understand user intent, enabling accessible and interactive communication
- Gesture-Based Volume Control: Designed a touchless volume control system using pinch gesture detection by tracking finger distance through webcam input, enabling real-time system interaction
- Virtual Drawing Application: Created an interactive drawing tool that allows users to draw on screen using index finger tracking, incorporating features like color selection and canvas control

Machine Learning Projects

- Implemented multiple end-to-end machine learning solutions involving supervised, unsupervised, and deep learning techniques, including data preprocessing, feature engineering, model training, evaluation, and visualization
- Food Delivery Time Prediction: Built regression and classification models (Linear & Logistic Regression) to predict delivery time and classify orders as fast or delayed; performed EDA, feature engineering, and derived business insights from real-world factors like traffic and weather
- Advanced Delivery Prediction (CNN & Multi-Model Analysis): Extended the delivery prediction system using CNNs and multiple ML models (Naive Bayes, KNN, Decision Tree) with cross-validation and ROC analysis for performance comparison and optimization
- House Price Prediction: Developed a Linear Regression model to estimate housing prices based on features like area and number of rooms; evaluated performance using MAE, RMSE, and R^2 metrics
- Customer Segmentation (K-Means Clustering): Applied unsupervised learning to segment customers based on income and spending behavior; used the Elbow Method and visualizations to derive actionable marketing insights

- Cat vs Dog Image Classification (SVM): Built an SVM-based image classifier using preprocessed image features; evaluated model performance using confusion matrix and analyzed limitations of traditional ML in computer vision tasks
- Global Pollution & Energy Recovery Analysis: Applied regression, classification, and Apriori algorithm to analyze relationships between pollution levels and energy recovery; generated insights using association rules and predictive models
- Pollution Classification using ML Models: Implemented Naive Bayes, KNN, and Decision Tree classifiers to categorize pollution severity levels; compared model performance using accuracy, precision, recall, and F1-score
- Deforestation Analysis using SVM: Developed an SVM model to predict deforestation trends based on environmental and economic factors; performed feature analysis and visualization to identify key contributors

Agentic AI for Indian Cuisine Analysis

- Developed an AI-powered system to identify food items from images and estimate calories and portion sizes using computer vision techniques
- Implemented an agentic AI workflow to process user input, analyze food data, and generate structured outputs with minimal human intervention
- Integrated image-based recognition with nutritional insights, enabling users to understand calorie intake and food composition
- Designed a scalable pipeline combining AI models, data processing, and response generation for real-time food analysis
- Demonstrated application of AI in healthcare and dietary monitoring use case

Cloud-Based Note-Taking Application

- Developed a scalable web application to create, store, update, and manage notes in real time using a serverless cloud architecture
- Implemented Google Cloud Firestore (NoSQL) for efficient data storage with real-time synchronization across devices
- Utilized asynchronous programming (Async/Await) to ensure smooth, non-blocking interactions and improved application performance
- Designed a responsive frontend using HTML, CSS, and JavaScript, and deployed the application via Firebase Hosting/CDN for global accessibility

Research Paper: Understanding Chatbots and Conversational AI

June 2025

- Published in International Journal of Modern Developments in Engineering and Science, Feb 2026
- Conducted a comprehensive survey on the evolution of conversational AI, analyzing rule-based, retrieval-based, and generative models including transformer-based architectures like BERT and GPT
- Explored core components such as Natural Language Understanding (NLU), Dialogue Management, and Natural Language Generation (NLG), highlighting their role in building intelligent dialogue systems
- Performed data-driven comparative analysis using Power BI, visualizing chatbot adoption trends, framework usage, and model performance across industries
- Analyzed key challenges including bias, hallucination, privacy risks, and ethical implications, and proposed future directions for responsible AI development

Research Paper: Agentic AI for Autonomous Data Analytics: A Theoretical Framework with Visualization-Driven Analysis

June 2026

- Published in International Journal of Innovative Research in Technology
- Proposed ADAS (Agentic Data Analytics System), a five-module theoretical framework enabling an AI system to autonomously execute the complete data analytics lifecycle — from natural language goal interpretation to finished report generation — without continuous human intervention
- Designed and specified five interdependent architectural modules: a Goal Interpreter, Hierarchical Planning Engine, Tool-Use Controller, Reasoning and Validation Layer, and Autonomous Reporting Engine — each described at sufficient technical depth to serve as a direct blueprint for prototype construction

- Developed a formal confidence scoring mechanism $C(o_i) = \alpha V_i + \beta R_i + \gamma H_i + \delta K_i$ for conflict resolution, a DAG-based task planning model, and a three-gate validation framework for self-correcting error recovery — grounded in LLMs, Retrieval-Augmented Generation, and multi-agent orchestration
- Conducted a comparative analysis of six existing AI tools (OpenAI Code Interpreter, AutoGen, Julius AI, Watson Analytics, LangChain) across eight capability dimensions, and proposed a ten-metric evaluation framework for future empirical validation of the system

Work Experience

Data Science Intern-Prodigy Infotech

JULY 2025

Applied predictive analytics techniques (regression, clustering, decision trees) and exploratory data analysis to identify customer behavior patterns. Built interactive dashboards and visualizations using Python and BI tools to communicate insights effectively.

Web Development Intern-Prodigy Infotech

JULY 2025

Built responsive web interfaces using HTML, CSS, JavaScript, and modern frameworks. Assisted in backend integration, debugging, and deployment of web applications using version control tools.

Java Developer Intern-Alfido Tech

SEPT 2025 -OCT 2025

Developed Java applications using OOP concepts, data structures, and frameworks such as Spring and Hibernate. Integrated databases and optimized application performance through systematic debugging.

Machine Learning Intern-Prodigy Infotech

JAN 2026

Working on data preprocessing, model building, & performance evaluation using Python & core ML algorithms.

Certificiates

Python Developer – Alfido Tech (Oct 2025): Gained hands-on experience in Python programming, including data structures, OOP concepts, and real-world problem solving.

Generative AI Fundamentals – Edubridge Learning (Sep 2025): Acquired foundational knowledge of Generative AI, LLM concepts, and AI-driven automation applications.

Power BI Expert Professional Certification – Alfido Tech (Aug 2025): Built interactive dashboards and data visualizations using Power BI, including data modeling and DAX fundamentals.

Web Development Certification – International Training Institute (Oct 2024): Developed responsive web interfaces using HTML, CSS, JavaScript, Bootstrap, and basic database concepts.

Artificial Intelligence with Machine Learning – Remarkskill Education (Jan 2024): Implemented Machine Learning algorithms and NLP basics using Python, including data preprocessing and model evaluation.

Python and Artificial Intelligence – IHFC, IIT Delhi (Jun 2023): Learned core AI concepts, Python programming, and introductory generative AI tools and workflows.

Machine Learning - Tutedude (Dec 2025): Worked with ML concepts such as regression, classification, and performance metrics to solve practical problems.

Completed AWS From Zero to Hero The Complete Guide - Udemy (Dec 2025): gaining hands-on knowledge of core AWS services, including EC2, S3, IAM, and cloud deployment concepts.

Cloud Foundations - AWS Academy Graduate (Dec 2025): Certification, covering cloud computing fundamentals, architecture, security, and best practices.

Leadership Roles

- Conducted a technical webinar on Java Fundamentals for junior students under the department committee, simplifying core concepts and promoting peer learning
- Led and organized a sketching competition as Event Head during the college cultural fest, managing event planning, coordination, and execution
- Demonstrated strong leadership, communication, and team management skills through successful event handling and audience engagement